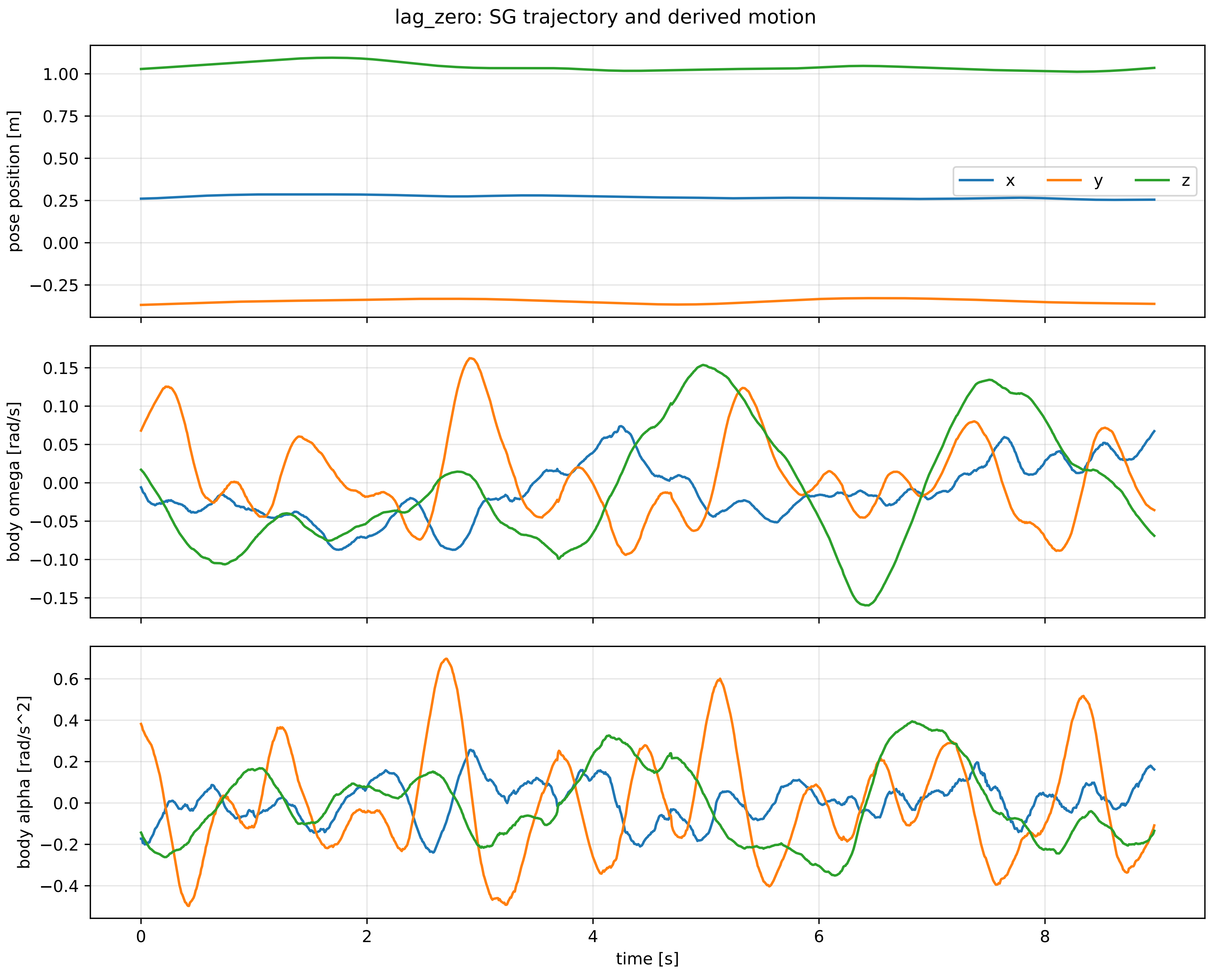
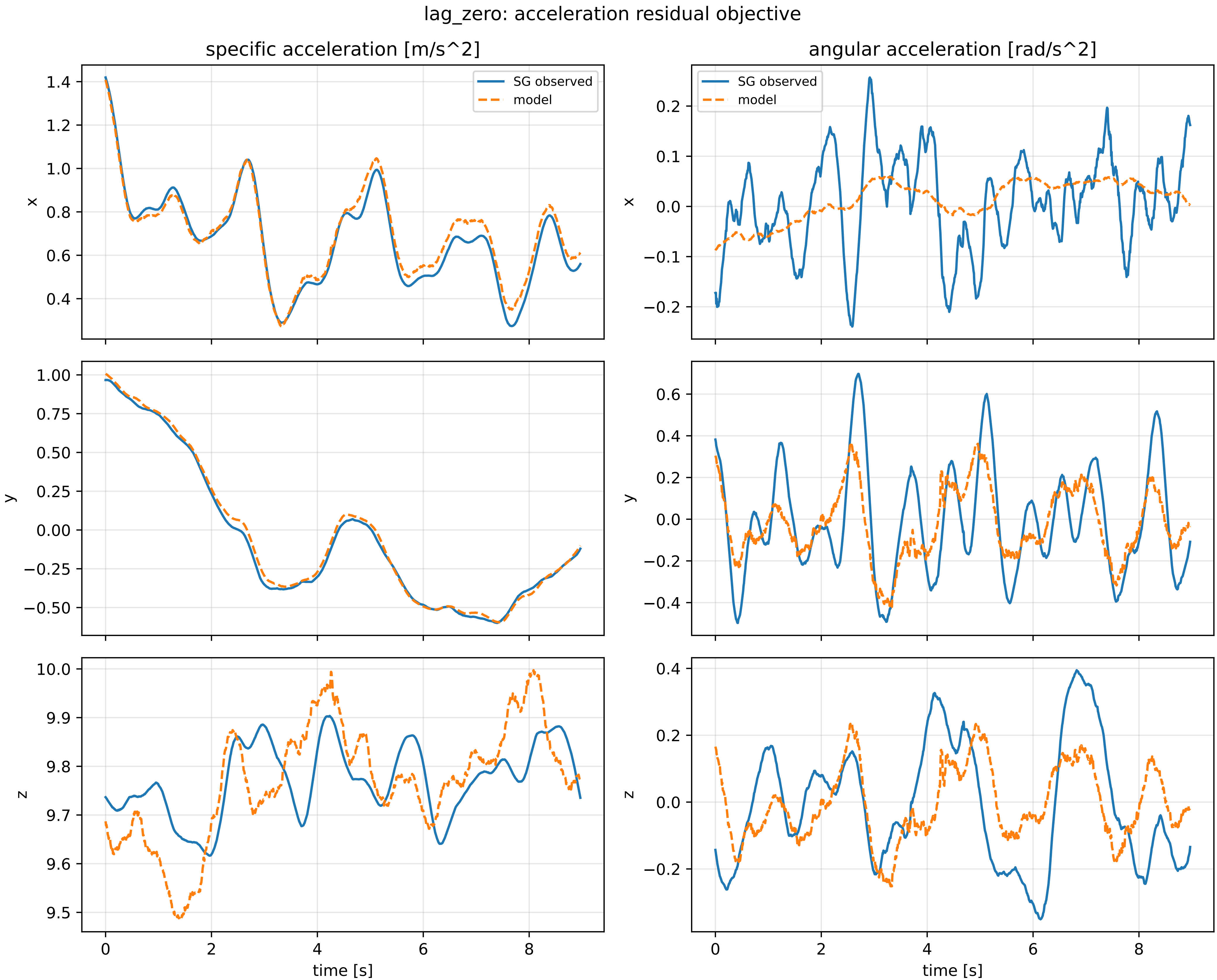
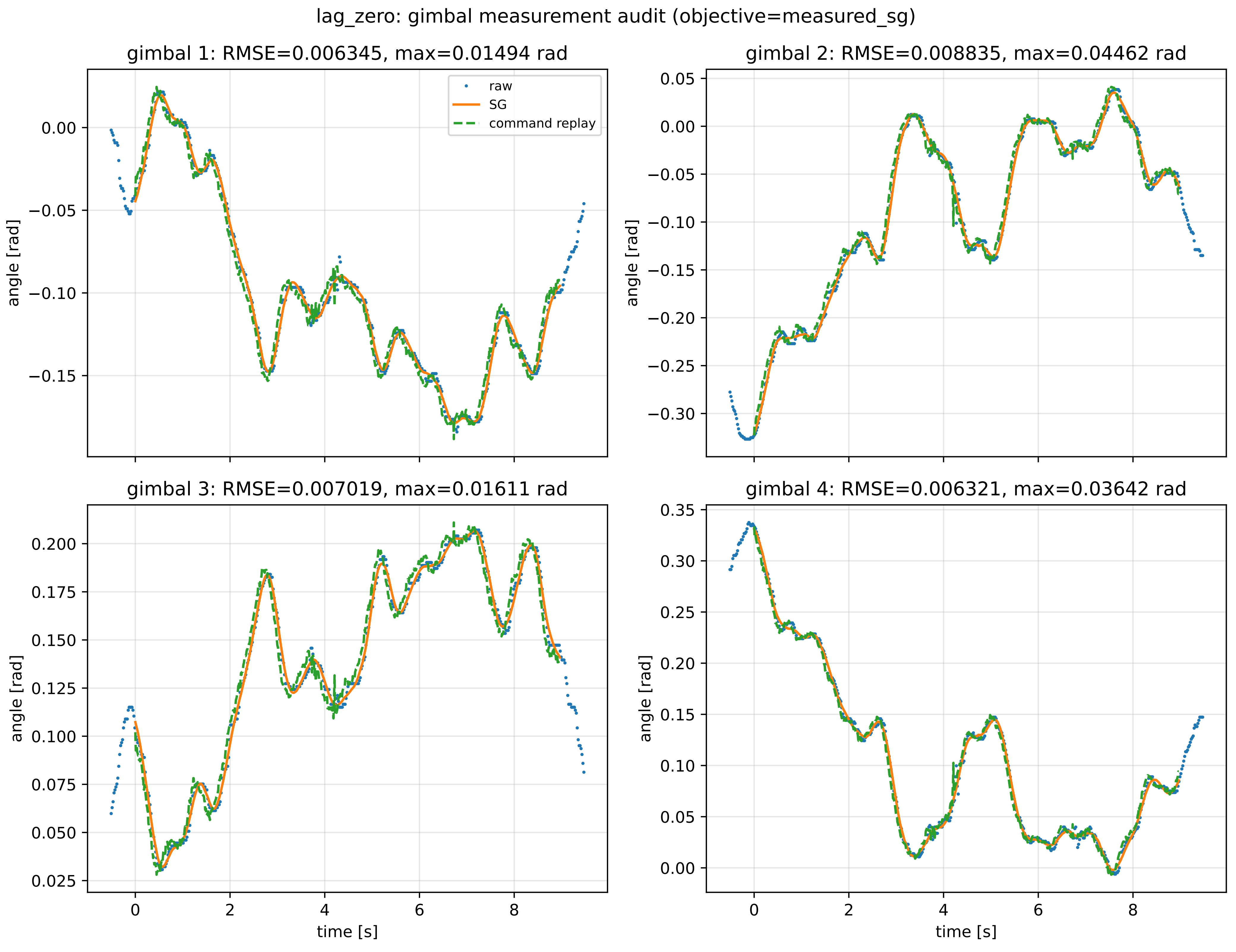


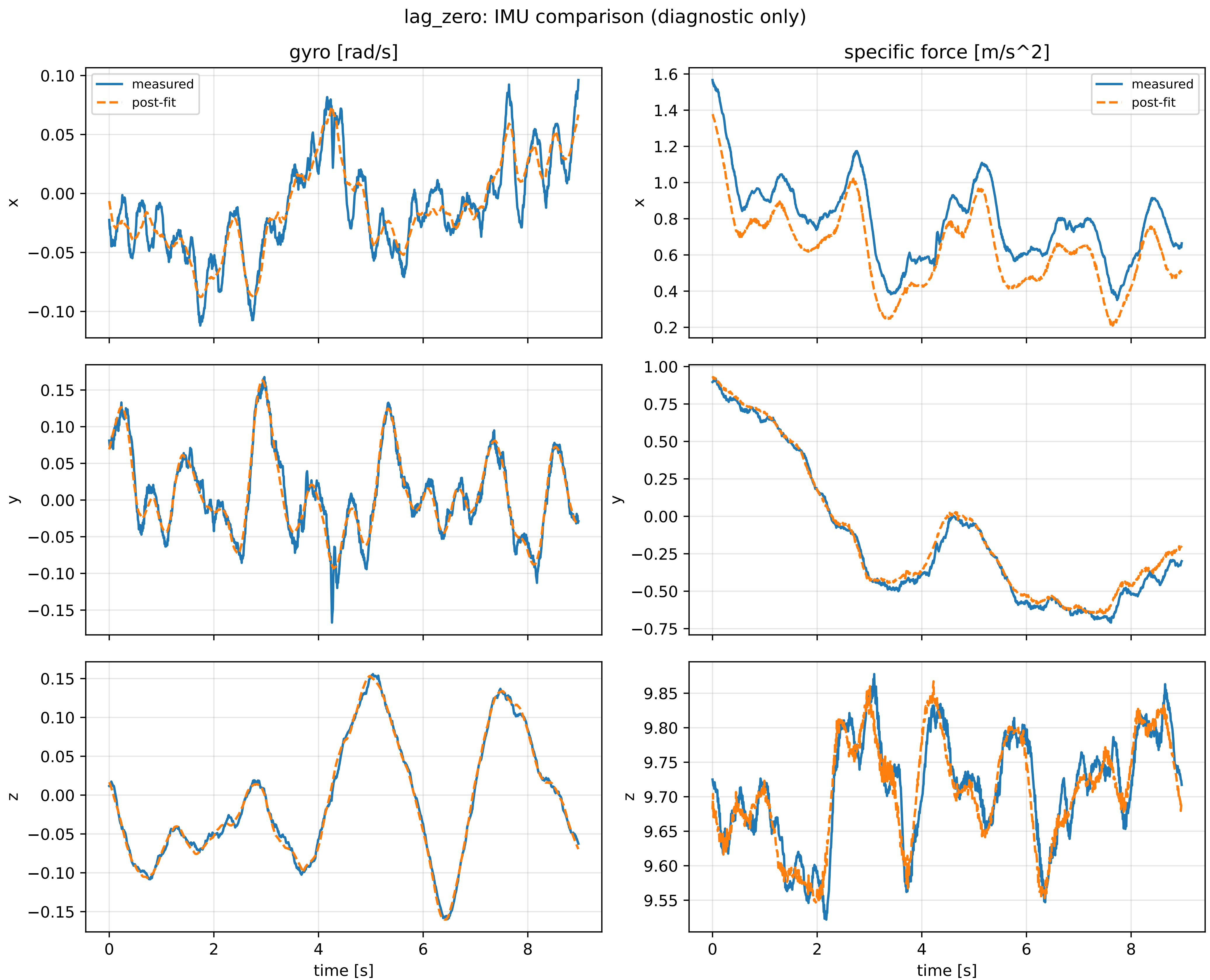
lag_zero: SG trajectory and derived motion
pose position [m]
1.00
0.75
0.50
0.25
0.00
−0.25
x
y
z
body omega [rad/s]
0.15
0.10
0.05
0.00
−0.05
−0.10
−0.15
body alpha [rad/s^2]
0.6
0.4
0.2
0.0
−0.2
−0.4
0
2
4
6
8
time [s]



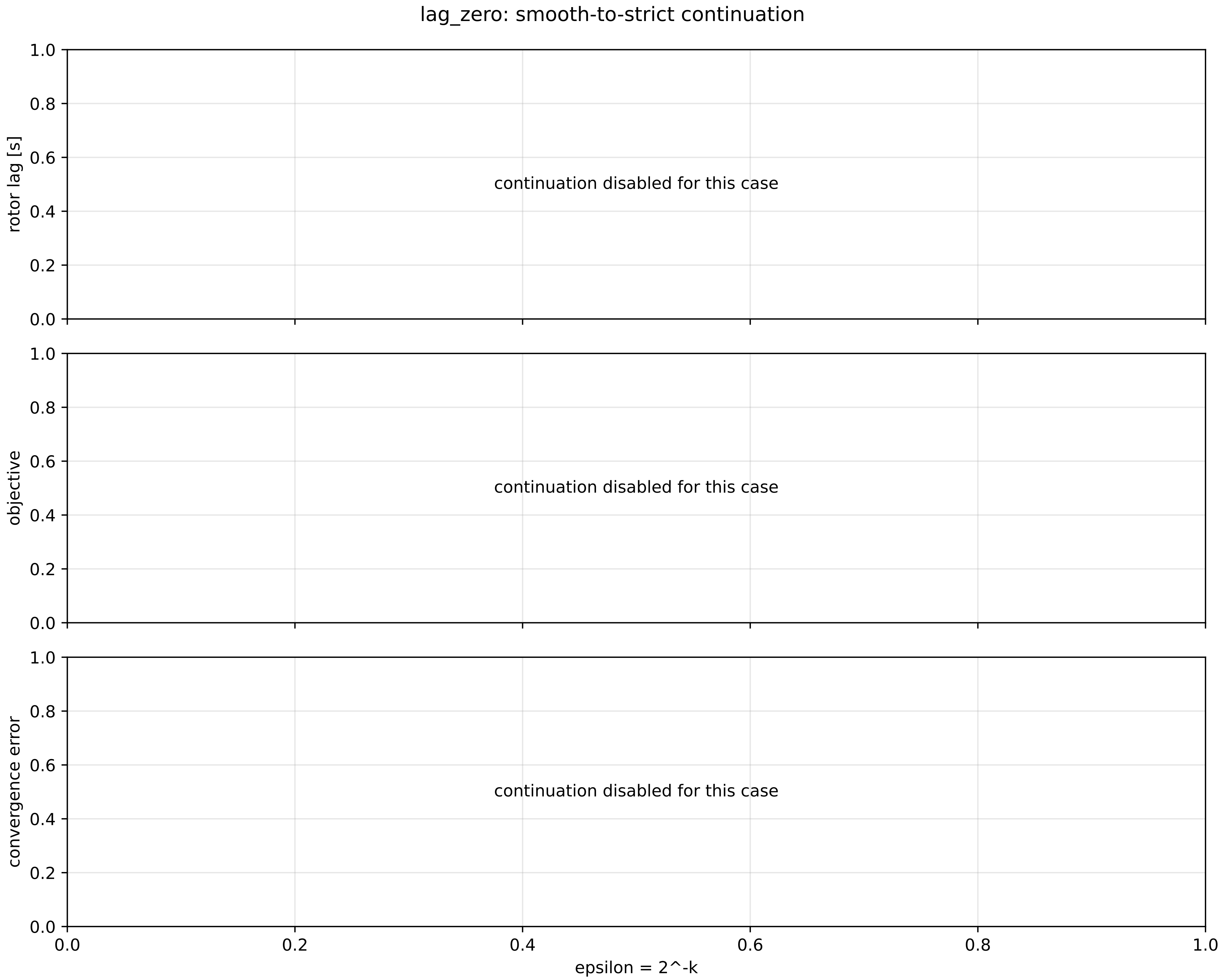
lag_zero: acceleration residual objective
specific acceleration [m/s^2]
angular acceleration [rad/s^2]
SG observed
model
x
y
z
time [s]



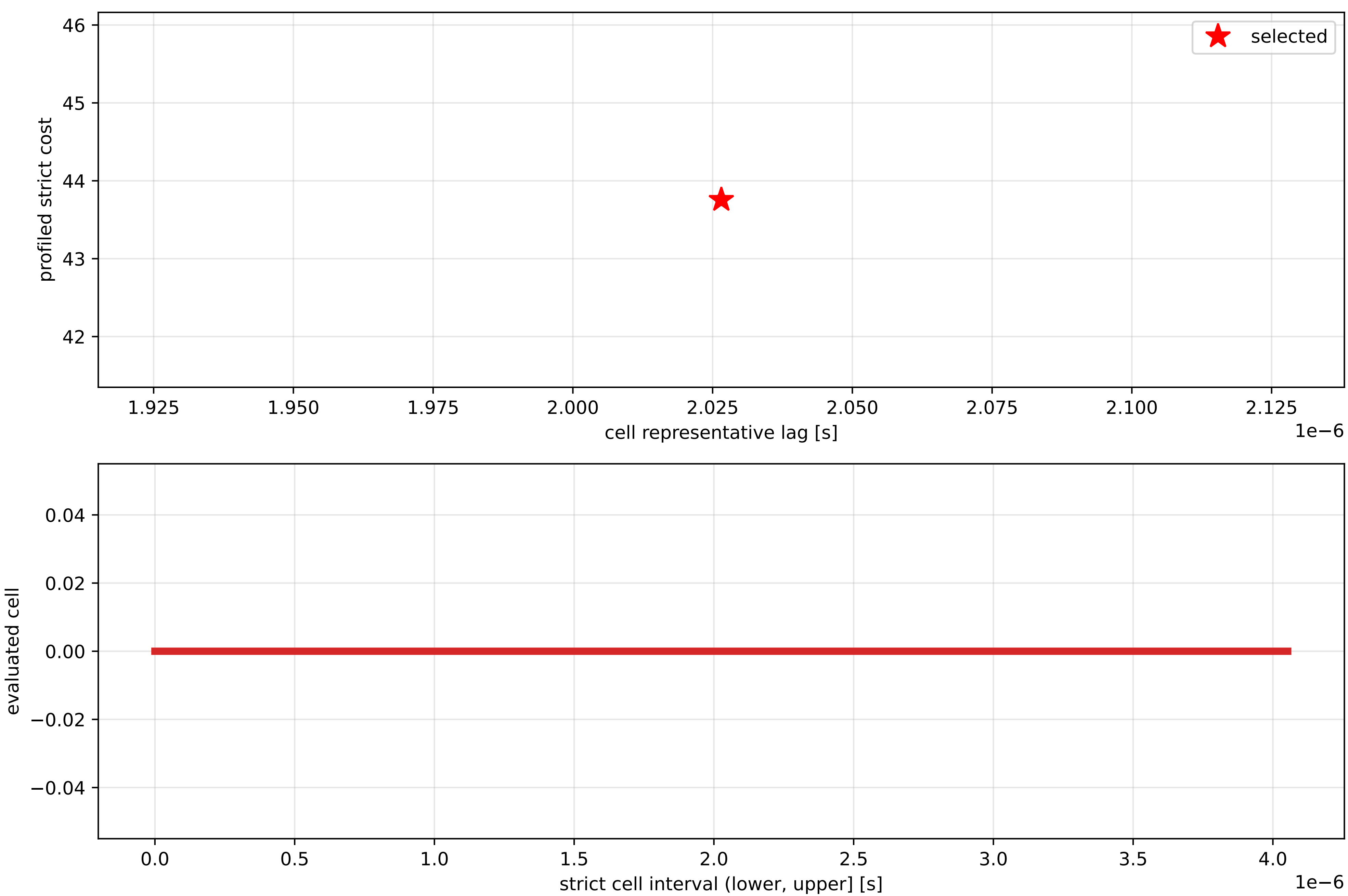
lag_zero: gimbal measurement audit (objective=measured_sg)
gimbal 1: RMSE=0.006345, max=0.01494 rad
gimbal 2: RMSE=0.008835, max=0.04462 rad
gimbal 3: RMSE=0.007019, max=0.01611 rad
gimbal 4: RMSE=0.006321, max=0.03642 rad
raw
SG
command replay
angle [rad]
time [s]



lag_zero: IMU comparison (diagnostic only)
gyro [rad/s]
specific force [m/s^2]
measured
post-fit
x
y
z
time [s]


lag_zero: smooth-to-strict continuation
rotor lag [s]
objective
convergence error
continuation disabled for this case
continuation disabled for this case
continuation disabled for this case
epsilon = 2^-k
0.0
0.2
0.4
0.6
0.8
1.0

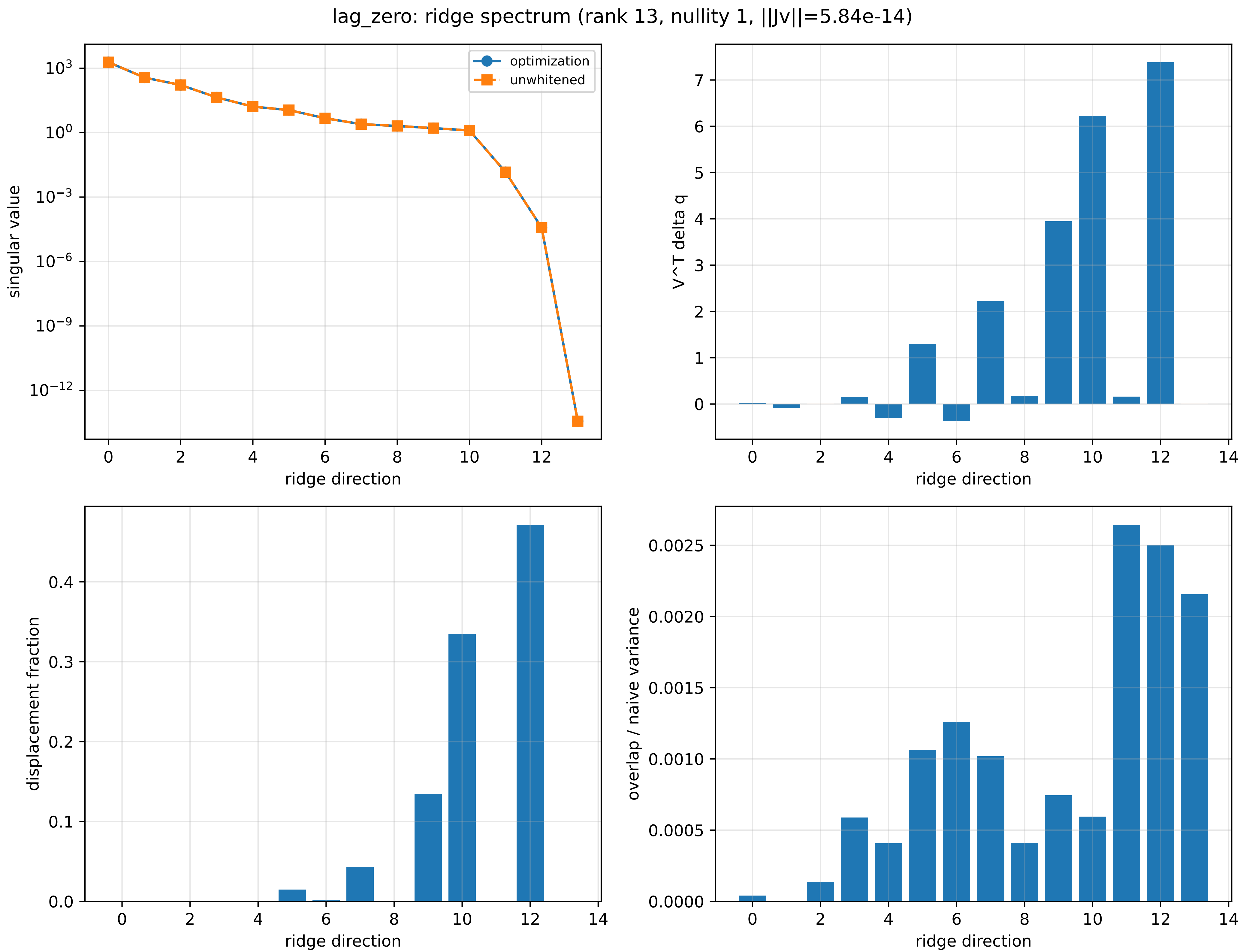
lag_zero: exact strict-ZOH lag cells

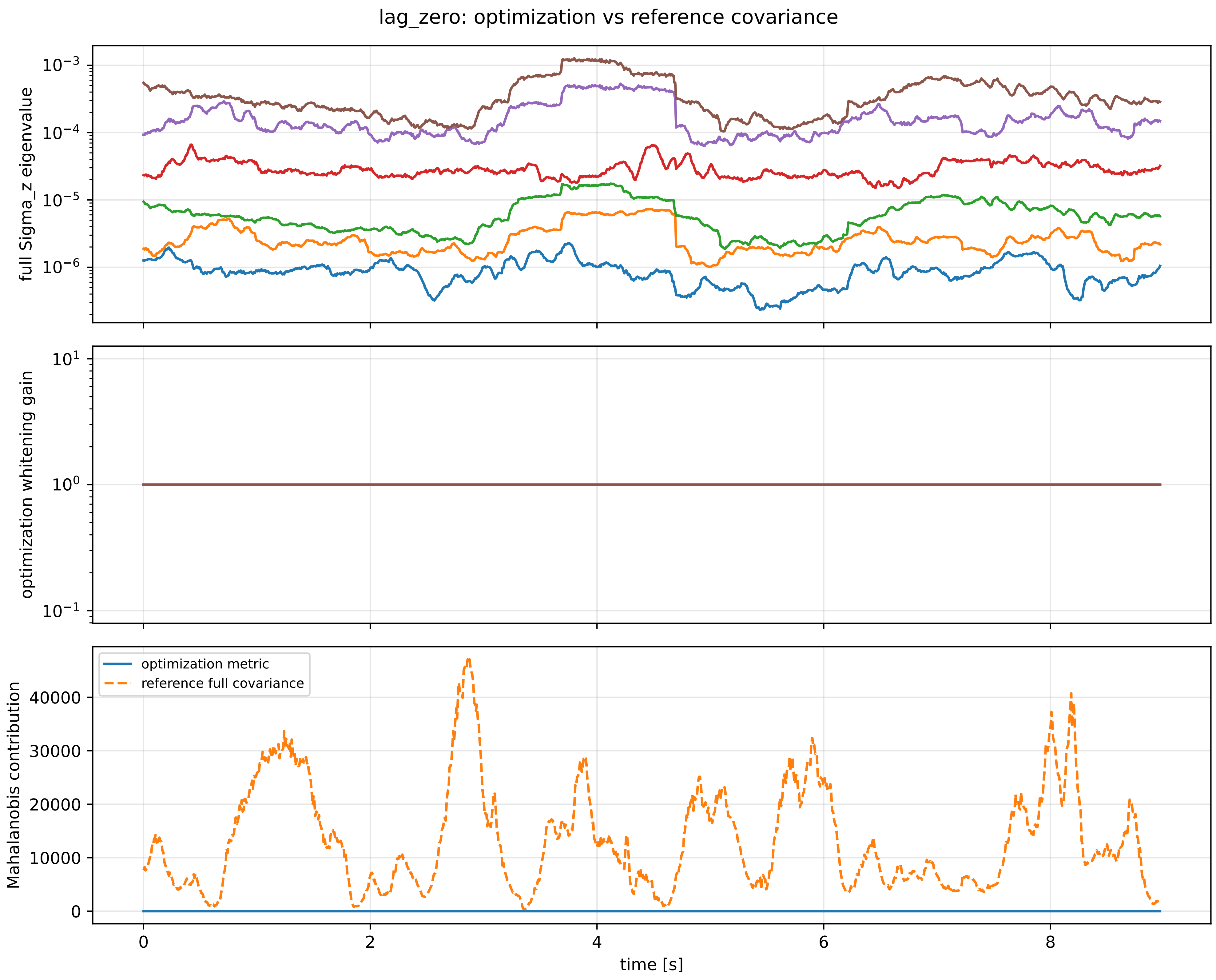



selected (0, 4.05311584e-06] s; final smooth 0 s; data boundary=False

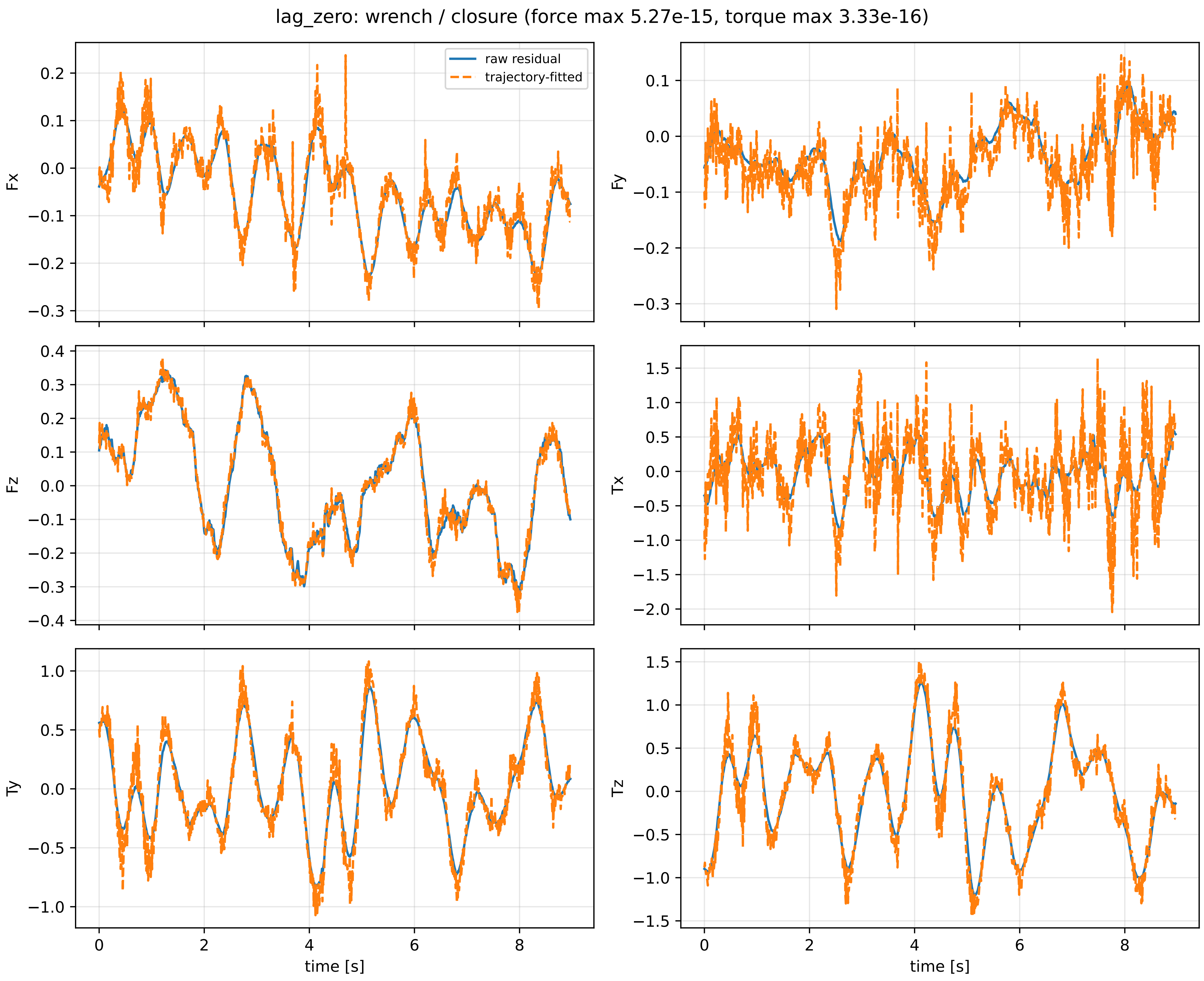
# lag_zero: parameter estimate

```
Case: lag_zero
status: completed (all stages completed)

mass [kg]: nominal 2.35159759; estimated 1.73084255
inertia [kg m^2]:
[[ 3.52748921  0.05023057  0.18147087]
 [ 0.05023057  1.27138599 -1.50478021]
 [ 0.18147087 -1.50478021  2.47772158]]
CoG body [m]: [-0.09942276 -0.05796506 -0.14849218]
force effectiveness: [0.9587045  0.54969294 0.21378805 0.64435307]

scale-free J/m [m^2]:
[[ 2.03801853  0.02902088  0.10484539]
 [ 0.02902088  0.73454745 -0.86939173]
 [ 0.10484539 -0.86939173  1.43151182]]
scale-free f/m: [0.55389469 0.31758691 0.12351675 0.37227712]

rotor lag cell: (0, 4.05311584e-06] s
representative: 0 s
```


lag_zero: ridge spectrum (rank 13, nullity 1, ||Jv||=5.84e-14)
optimization
unwhitened
singular value
ridge direction
V^T delta q
displacement fraction
overlap / naive variance



lag_zero: optimization vs reference covariance
full Sigma_z eigenvalue
optimization whitening gain
Mahalanobis contribution
optimization metric
reference full covariance
time [s]



lag_zero: wrench / closure (force max 5.27e-15, torque max 3.33e-16)
raw residual
trajectory-fitted
Fx
Fy
Fz
Tx
Ty
Tz
time [s]
time [s]